



“IN FUTURE YOU COULD BE FLYING FROM LONDON TO SYDNEY IN THIRTY MINUTES.”

Stephen Attenborough

ation. His greatest concern when he was making this was not that he wouldn't make it, but that it could just be the false dawn, that the spaceship would be taken to the Smithsonian Museum, Washington DC, where it now hangs. What was needed was an entrepreneur who could recognise the potential of this development. Now I don't think

there are many people who would probably better qualify to do something with it than Richard Branson, certainly in his attitude to adventure, to risk and to entrepreneurship. In 1999, he had announced we would be flying people to space within 2009 despite not knowing how to do it.

We had about 90,000 people who registered on our website to say they would love to get into space. These people come from 27 different countries across the world of which 80 per cent are men and 20 per cent women. The age range we have is 22 to 88. But many others want to make what we are doing possible by putting their money down, by getting the experience and enabling us to radically transform the face of space access for the future. And one of the possible developments again could be flying London to Sydney in probably half an hour. So, the future is bright. It is a new age of discovery.

D I S C U S S I O N

Q. I heard you say that it will be five minutes of excitement in space. Can it be more than that?

Attenborough: You get a lot more than five minutes of excitement. The actual time in weightlessness on this flight is about four-five minutes and that's determined by the height, the altitude that we fly, which in turn is determined by the amount of power that's required by the spaceship. I always equated it a little bit to people spending a similar amount of money preparing and climbing Mount Everest. Very few people have experienced this in their lifetime. The other thing that we have to remember is that this is the first step and we must demonstrate safety and commercial viability of a space flight for ordinary people. If we can do that we expect a whole lot of private sector money to come

into this industry because it looks attractive and we can see things develop quickly. So we can extend to orbital spacecraft a similar level of safety and the idea of a space hotel and a fortnight in space becomes a reality.

Q. During re-entry there is so much of resistance from the atmosphere that the temperature goes up very high. In the absence of heat shields, don't you have any special treatment on the outer skin of the aircraft?

Attenborough: You see if you look at the model, there is very limited film protection and there is no special in-shield that is needed. And there are two reasons for this. First, remember we are not going into the orbit with this space craft. The main reason that we do not need this type of protection is because the aerodynamic design and the rotated

wings, the feathering effect, actually slows the spacecraft in the upper atmosphere to an extent where you never build up excessive heat. So the carbon fibre construction with the exception of a couple of specific areas is robust enough to withstand the heat that is built up in that feathered re-entry. And that is something which is just a masterpiece of design and something which really transforms the level of safety.

Q. There have been ethical issues about stem cell research in the US where people say that life starts at conception rather than at birth and if you are using embryonic stem cells you are violating life. Why is there so much controversy over it and what can you do to convince people that it is good?

Saldanha: Since you are using

live embryos as a source of stem cell, there always will be ethical issues especially with certain conservative governments. There is a lot more acceptance, of late, in terms of patenting of stem cells and patenting of speciality embryonic stem cells as compared to the past. I know in India there are a couple of institutions which have ongoing efforts in embryonic stem cells. Though they are still in their early stages but some of the work which has been done is extremely exciting. There is clearly an ethical issue—it depends on which way you look at it. If you end up saving life and are actually able to treat degenerative diseases, there is a huge argument for allowing embryonic stem cells to be used.

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